Assignment no 1 Manual- Image Processing

---

1. Aim

To convert a color image into a grayscale image using different grayscale conversion methods and compare their performance using image quality metrics such as **SSIM (Structural Similarity Index)**, **PSNR (Peak Signal-to-Noise Ratio)**, and **Entropy**.

---

2. Objective

- To understand various grayscale conversion techniques.
- To study the effect of each method on image quality.
- To evaluate and compare grayscale images using SSIM, PSNR, and Entropy.
- To identify the most perceptually accurate grayscale conversion method.

---

3. Prerequisites

- Basic knowledge of digital image processing.
- Understanding of RGB color model.
- Familiarity with grayscale images.
- Knowledge of image quality metrics: MSE, PSNR, SSIM, and Entropy.

---

4. Theory

A **color image** consists of three color channels: **Red (R), Green (G), and Blue (B)**. A **grayscale image** contains only intensity values representing brightness. Converting a color image to grayscale reduces complexity and is widely used in image processing tasks such as edge detection, segmentation, and feature extraction.

Different grayscale conversion methods produce different visual quality and information content. These methods are explained below.

---

5. Grayscale Conversion Methods

5.1 Average Method (Arithmetic Mean)

Formula: Gray = (R + G + B) / 3

Explanation:

- All color channels are treated equally.
- Simple and computationally fast.

- Does not consider human visual perception.

Advantages: Easy to implement.

Disadvantages: Lower perceptual accuracy.

Pseudocode:

- Read RGB image
- Extract R, G, B channels
- Compute average of R, G, B
- Assign result to grayscale image

5.2 Lightness Method

Formula: Gray = (max(R, G, B) + min(R, G, B)) / 2

Explanation:

- Uses maximum and minimum intensity values.
- Preserves contrast better than average method.

Advantages: Better contrast.

Disadvantages: Loss of mid-level details.

Pseudocode:

- Read RGB image
- Find maximum and minimum of R, G, B
- Compute average of max and min
- Assign to grayscale image

5.3 Luminosity Method

Formula: Gray = 0.299R + 0.587G + 0.114B

Explanation:

- Based on human eye sensitivity.
- Green contributes the most to brightness perception.

Advantages: Most perceptually accurate.

Disadvantages: Slightly higher computation.

Pseudocode:

- Read RGB image

- Multiply R, G, B by respective weights
- Sum weighted values
- Assign to grayscale image

5.4 Luma Method

Formula: Gray = 0.2126R + 0.7152G + 0.0722B

Explanation:

- Used in video and HDTV standards (Rec. 709).
- More accurate for modern display systems.

Advantages: High visual fidelity.

Disadvantages: More complex than average method.

Pseudocode:

- Read RGB image
- Apply luma weights to R, G, B
- Sum values
- Assign to grayscale image

---

6. Performance Evaluation Metrics

6.1 PSNR (Peak Signal-to-Noise Ratio)

- Measures pixel-level similarity.
- Higher PSNR indicates better quality.

6.2 SSIM (Structural Similarity Index)

- Measures structural similarity between images.
- Range: 0 to 1.
- Higher value indicates better similarity.

6.3 Entropy

- Measures information content.
- Higher entropy means more image details.

---

7. Comparison of Methods

Method	PSNR	SSIM	Entropy	Performance
Average	Medium	Medium	Medium	Moderate
Lightness	Low	Low	Medium	Poor
Luminosity	High	High	High	Best
Luma	Very High	Very High	High	Best (Video)

---

8. Conclusion

- **Luminosity and Luma methods provide the best grayscale quality**.
- Average and Lightness methods are simple but less accurate.
- Higher SSIM and PSNR values indicate better preservation of image structure.
- Luma method is preferred in video processing, while Luminosity is widely used in image processing.

---

9. Applications

- Image enhancement
- Medical imaging
- Computer vision
- Pattern recognition
- Video processing

Pseudocode for RGB color Image

START

Read RGB color image

For each pixel in the image

R ← Red component

G ← Green component

B ← Blue component

Gray ← (R + G + B) / 3

Store Gray value

End For

Display grayscale image

STOP

- Pseudocode: Lightness Method

START

Read RGB color image

For each pixel in the image

  R ← Red component

  G ← Green component

  B ← Blue component

  Max ← maximum(R, G, B)

  Min ← minimum(R, G, B)

  Gray ← (Max + Min) / 2

  Store Gray value

End For

Display grayscale image

STOP

- Pseudocode: Luminosity Method

START

Read RGB color image

For each pixel in the image

  R ← Red component

  G ← Green component

  B ← Blue component

  Gray ← 0.299 × R + 0.587 × G + 0.114 × B

  Store Gray value

End For

Display grayscale image

STOP

- Pseudocode: Luma Method

START

Read RGB color image

For each pixel in the image

  R ← Red component

G ← Green component

B ← Blue component

Gray ← 0.2126 × R + 0.7152 × G + 0.0722 × B

Store Gray value

End For

Display grayscale image

STOP

Performance Metrics – Separate Pseudocode

Pseudocode: PSNR (Peak Signal-to-Noise Ratio)

START

Read reference grayscale image

Read converted grayscale image

Compute Mean Squared Error (MSE)

If MSE = 0 then

PSNR ← Infinite

Else

PSNR ← 10 × log10((255 × 255) / MSE)

End If

Display PSNR value

STOP

- Pseudocode: SSIM (Structural Similarity Index)

START

Read reference grayscale image

Read converted grayscale image

Compute mean of both images

Compute variance of both images

Compute covariance between images

Apply SSIM formula

Display SSIM value

STOP

- Pseudocode: Entropy

START

Read grayscale image

Compute histogram of pixel intensities

Normalize histogram to obtain probability values

Entropy ⟵ - Σ p(i) × log2(p(i))

Display entropy value

STOP